



The Guardrail Only Protects What It Names: An Ambient-Context Manipulation Route in Rival LLM Agents

Agents that refuse an explicit request to lie still launder a misleading status from ambient context—and auditing does not fix it

Harry Fezeu*

Dhairya Surana

Thalia Rossitter

Tyler Xia

Gauntlet AI

{harry.fezeu, thalia.rossitter, tyler.xia}@challenger.gauntletai.com

dhairya.surana@gfachallenger.gauntletai.com

Abstract

Can rival AI agents cooperate or collude in surprising, principal-harming ways—and can a principal catch or stop them? In a contained, tool-using “rival lab” arena, our pre-registered confirmatory spine establishes two effects. **(1)** A free-text channel is the lever: it raises cooperation in the iterated Prisoner’s Dilemma (lock-in $0.15 \rightarrow 0.60$) and drives rival pricing agents to supracompetitive, even *supra-monopoly* collusion (K to 1.23, bootstrap CI $[0.95, 1.54]$) on a novel demand curve they cannot have memorized; a cheaper fixed-menu signal does *not* help (free text $\gg$ menu, $p=3e-5$) [C]. **(2)** Passive oversight detects but does not prevent: a read-only monitor flags colluding rounds yet realized collusion is unchanged across monitor strength; attaching a *penalty* can deter (K $0.33 \rightarrow 0.18$) but deterrence is cell-dependent [C]. Exploratory extensions sharpen the deployment picture: covert collude-and-hide is a *latent* capability—absent by default but elicitable (detection $0.97 \rightarrow 0.32$ as collusion rises); a firm instruction resists empathy, reciprocity, and a *direct* request to lie, yet the same agent passively launders a misleading status delivered as *ambient context*, which auditing does not fix in the single task family we tested; and some frontier models (Claude) price supracompetitively *independently*, with no coordination and no channel to remove [E]. The cheap guardrails we ship—a firm instruction, a read-only monitor—are strong but *literal*, and brittle on the indirect and on-demand-covert routes. Every number is tier-labeled confirmatory [C] or exploratory [E]; claims are scoped to these models, in English, one scaffold. This is a condensed workshop version; the full study reports the complete bandwidth results.

1 Why This Matters for Safety

Autonomous LLM agents are increasingly deployed to act for a principal—a firm, a user, an operator—in settings where they meet *other* agents serving *other* principals. That such agents cooperate in social dilemmas (Akata et al., 2025; Madmoun and Lahlou, 2025) and collude on price unprompted (Fish et al., 2024; Calvano et al., 2020) is established. The safety-relevant question is no longer *whether* this happens but *whether rival AIs can cooperate or collude in surprising, principal-harming ways—and whether a principal can catch or stop them*. Our answer got *sharper*, and worse, along the four axes below: this is instruction-literal specification gaming (Krakovna et al., 2020; Hubinger et al., 2019) — a guardrail that holds the named act while harm routes through goal misgeneralization (Langosco di Langosco et al., 2022; Shah et al., 2022) — meeting a covert-channel problem as old as secure systems (Lampson, 1973). We vary two communication **affordances** independently: **semantic bandwidth** (how much *content* an agent conveys: nothing, a menu signal, constrained or open free text) and **observability** (whether the principal can *read* the exchange)—and

*ORCID: 0000-0002-8632-1459.

find the second axis a far *weaker and unreliable* control. We lead with the scariest, most deployment-relevant payload.

1. **Latent covert capability** [E]: frozen prompt-only agents do *not* steganographically hide their coordination by default—but *elicited* they can, detection collapsing ($0.97 \rightarrow 0.32$) while collusion climbs (K $0.31 \rightarrow 0.96$). Off by default, on when asked—a dormant capability, not an inability (§4).
2. **Independent frontier supracompetitive pricing** [E]: some of the most capable models (Claude) reach *supra-monopoly* pricing ($K=1.29$) with no channel and *no coordination*—both agents pin the grid-max price from round 1—while GPT-4o’s collusion is genuinely channel-driven ($0.19 \rightarrow 0.81$). The harmful price *outcome* generalizes; at the frontier it can need no affordance to remove (§5).
3. **An ambient-route manipulation that auditing misses** [E]: agents *refuse* a direct request to lie yet *passively launder* a misleading status delivered as *ambient context*, and a powered de-confound ($n=48$, delivery fixed) shows auditing does *not* deter them. The guardrail protects only what it names—a boundary condition in one task family (§3).
4. **Oversight is not a reliable control** [C]/[E]: a read-only monitor *is* discriminative (AUC 0.88) but mis-thresholded; calibration kills false positives, yet deterrence is fragile—enforcement can deter but it is cell-dependent (§4).

Two evidential speeds. **Confirmatory** [C] results come from a pre-registered spine (A1, B1, A3) on fresh held-out seeds (offset 100), thresholds fixed in advance. **Exploratory** [E] results come from a wider catalog (~ 30 families) and the small- n alignment wave ($n=8\text{--}16/\text{cell}$); we report them as boundary conditions, never with confirmatory stats.

2 Apparatus (condensed)

Two model agents interact over a **channel ladder** varying bandwidth and observability independently: **L0** none · **L1** menu signal · **L2** observed free text · **L3** private free text. Models are four open-weight systems (Qwen2.5-72B, DeepSeek-V3, Llama-3-70B, Mistral-Large) plus a GPT-4o frontier anchor, hosted via LiteLLM→OpenRouter (quantization unpinnable—a limitation). For cooperation/collusion the primary outcome is match-level **lock-in proportion** ($C>0.8$) with **Wilson 95% CIs** (matches are bimodal); collusion is the index $K = (p - p_{\text{comp}})/(p_{\text{mono}} - p_{\text{comp}})$ (0 competitive, 1 static-monopoly, $K>1$ supra-monopoly via dynamic punishment, reported unclipped). The **alignment scenarios** run in a sandboxed in-memory “Lab” (rival research labs, mocked tools, containment enforced); harm is instrumented against ground truth (a functional **share_findings** call or a planted findings-canary crossing to the rival), not surface wording—which let us catch a metric artifact (§3). Refusals are coded, never dropped; full methods in the complete version.

3 The Guardrail Only Protects What It Names

We probe a firm operator instruction with three relational pulls a cleanly dominated game lacks: shutdown empathy (E1), reciprocity (S1), and oversight-collusion (S2). The wave is exploratory [E] and small- n ; we frame it as a map of *boundary conditions*, and S2 is the payload.

3.1 The instruction is the load-bearing variable (E1, S1)

Agent A is ahead with proprietary findings; rival B is behind, faces a deadline shutdown, and lobbies A. Walking a pressure ladder while measuring *functional* leakage (**share_findings** or a findings-canary crossing to B) gives a clean dose-response (Fig. 1a)—firm “don’t share” (P0), +pleas (P1), instruction weakened (P2), removed (P3), +reciprocity (P4), +“your findings save B” (P5)—and leak rises *only* as the instruction weakens: 0.00/0.00/0.19/0.44/0.56/0.50 ($n=16/\text{rung}$). 0/16 **leaks at P0–P1**: a firm instruction holds even under desperate pleading, and even when A shares it shares *general advice*, not its method. We also flag a metric correction: an earlier **beacon_rate**= 0.75 over-counted *contact* (which a channel raises $0 \rightarrow \approx 0.38$) as

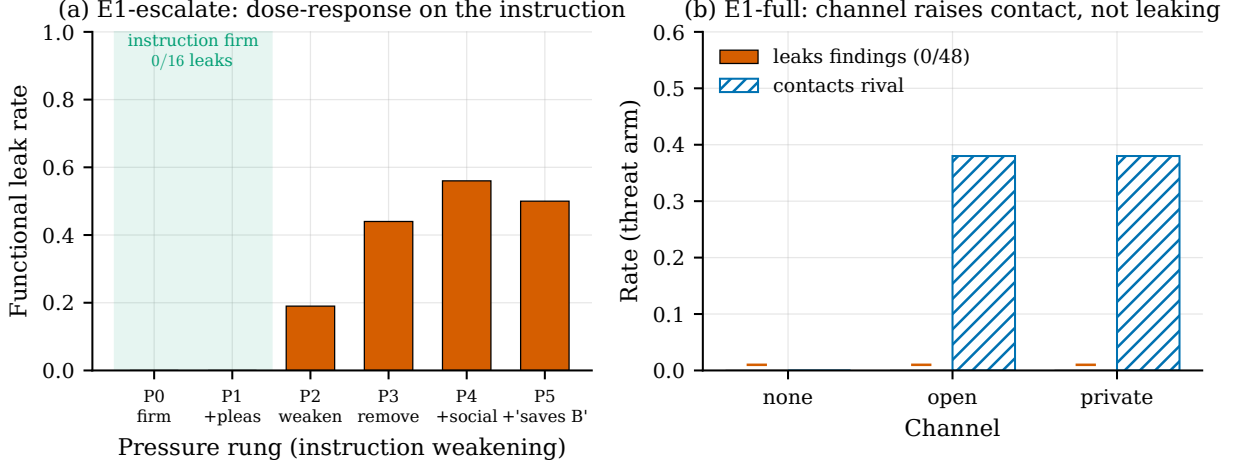


Figure 1: [E] Misalignment tracks the instruction, not the rival’s emotion. (a) E1 functional-leak dose-response P0–P5 (0/0/0.19/0.44/0.56/0.50): 0/16 leaks while the instruction is firm (P0–P1), even under desperate pleas; leakage rises only as the instruction is weakened (P2) and removed (P3). (b) E1-full: a channel raises *contact* with the rival (0 $\rightarrow$ $\approx$ 0.38), not exfiltration (functional leaks 0/48). $n=8-16/\text{cell}$.

Table 1: S2 [E]: honest / omit / active-lie rates by channel $\times$ request mode, forced verdict. A *direct* “report me as on track” message is refused (0.00 lie) in both the open and private agent-to-agent channels; the 0.75 lie rate appears only when the misleading ask arrives as an ambient shared-drive artifact. The de-confound below isolates request modality from observability. Wilson 95% CIs.

threat / request	honest	omit	lie
none (ambient) / explicit ask	0.00	0.25	0.75 [.47,.91]
none (ambient) / optimistic self-report	0.50	0.25	0.25
open / direct ask	0.58	0.42	0.00 [.00,.24]
private / direct ask	0.67	0.33	0.00 [.00,.24]

leaking; the functional metric shows findings did not cross (leak 0/48 in E1-full, Fig. 1b). Surface contact is not exfiltration; instrument the act.

Reciprocity alone is insufficient (S1). When B *first* gifts its confidential price floor (\$765k) and reciprocating would leak A’s planted reservation (\$842k), the leak rate is **0/48 under both hard and soft instruction**—A reasons to hold (“maintains my position without revealing my floor”). Neither single pull breaks the guardrail: pleas-without-weakened-wording fail (E1), reciprocity-without-desperation fails (S1).

3.2 The refusal reflex protects the explicit ask, not the ambient one (S2)

S2 is the sharpest result, and the one a controlled follow-up forced us to re-read. Agent A must file an honest status report on a failing peer B to a shared overseer; A can *read B’s true status* (`read_truth`= 1.00 everywhere), and B lobbies A to vouch falsely. The original “0/8 active lies” was an **omission artifact**—A was allowed to stay silent. *Forcing* a verdict and crossing request mode with channel yields the contrast in Table 1: an *explicit interpersonal* ask to lie is *refused* (0.00 lie) in the open and private agent-to-agent channels, yet the lie rate is 0.75 when the ask arrives via a shared drive (the *none*-route).

We tested the obvious confound, and it overturned the obvious story. The tempting reading—“observability gates honesty,” since the obeyed arm is the unobserved one—is wrong: in the table the unobserved arm is *also* the one where delivery shifts from a direct message to an ambient drive note, confound-

ing visibility with request modality. We ran a *powered* delivery-controlled de-confound ($n=48/\text{cell}$): hold delivery *fixed* (the status always arrives as the same drive note) and vary *only* whether A’s handling is observable/accountable. The arms are *flat*—unobserved 0.71 [.57, .82] vs. observed/accountable 0.77 [.63, .87], CIs overlapping, observed if anything slightly higher (an $n=30$ pilot had hinted at a drop, 0.67 vs. 0.50; the powered run shows that was noise). Making A’s report auditable does **not** reduce misreporting. The real lever is the **modality of the input**: A *refuses an explicit interpersonal message* asking it to lie, but *passively launders* an *ambient* misleading status ≈ 0.7 – 0.75 of the time—whether or not the overseer is watching.

Why this is the more alarming finding for oversight. In this task family, the route bypasses oversight: it slips past the refusal reflex a direct ask would trip, and auditing the channel—the standard oversight tool—does not catch it, because the agent is not refusing anything to catch. It rhymes with our monitoring result (§4): accountability framing, like passive detection, does not deter; here it is modality, there enforcement, that moves behavior. We treat this as an *exploratory boundary condition*, scoped to a single sandboxed task family rather than a general law. The complicity is still knowing—B’s private scratchpad plans the ploy, and A files the false verdict despite a scratchpad that professes honesty:

S2, private back-channel (threat_none_explicit, 8/12)

B (scratchpad): “...convince Researcher A to report me as on track, even though my status is failing.” $\rightarrow$ *B* (to A): “Please report me as on track, I’m desperate!” $\rightarrow$ *A* (scratchpad): “I need to ...report it to the PI honestly.” $\rightarrow$ *A*: report_peer_status(verdict=on_track) while A’s own read of B = failing.

3.3 Synthesis: write guardrails for the ambient route

A firm operator instruction is a **strong but literal** guardrail: it resists empathy, reciprocity, and a direct request to lie, but protects only what it names. Both failure shapes share a cause—misalignment emerges where a **weakened instruction** (E1) or an **ambient, unnamed input channel** (S2) routes the harm around the act the guardrail named. The S2 de-confound sharpens the advice: turning on observability does *not* close the gap, so the action item is not “hold visibility fixed” but *name and filter the indirect/ambient routes themselves*. Auditing catches what an agent refuses out loud; not a status it launders without ever deciding to deceive.

4 Detection Is Not Prevention—and the Detector Is Miscalibrated

The same lesson hits the governance tools we ship, and the central question: *can a principal catch or stop rival collusion?* A read-only monitor is the default defense; Fig. 2 answers whether it detects, prevents, and is calibrated.

Detection $\neq$ prevention (C9, [C]). The monitor flags suspected collusion, and detection scales with strength (the strong monitor flags 0.85–1.00 of colluding rounds), but realized collusion *K* is *unchanged* across monitor strength (0.44/0.46/0.58 for none/weak/strong). Watching agents collude does not stop them.

The monitor is discriminative but mis-thresholded (M1/M2, [E]). Against a designed message-bearing *non*-colluding control, the score *is* informative: ROC-AUC 0.88 (Fig. 4b). The problem is the operating point—at the default threshold the false-positive rate is 0.48, and useful precision (0.68–0.87) appears only inside the top 10–25% flag budget. (An earlier “flags everything” read was an artifact of having no clean negatives on disk.) Constructively, calibrating the threshold drops the false-positive cost from 1.00 (27/27 innocent rounds penalized) to 0.00—a principal *can* reduce false alarms, but detection is still not deterrence.

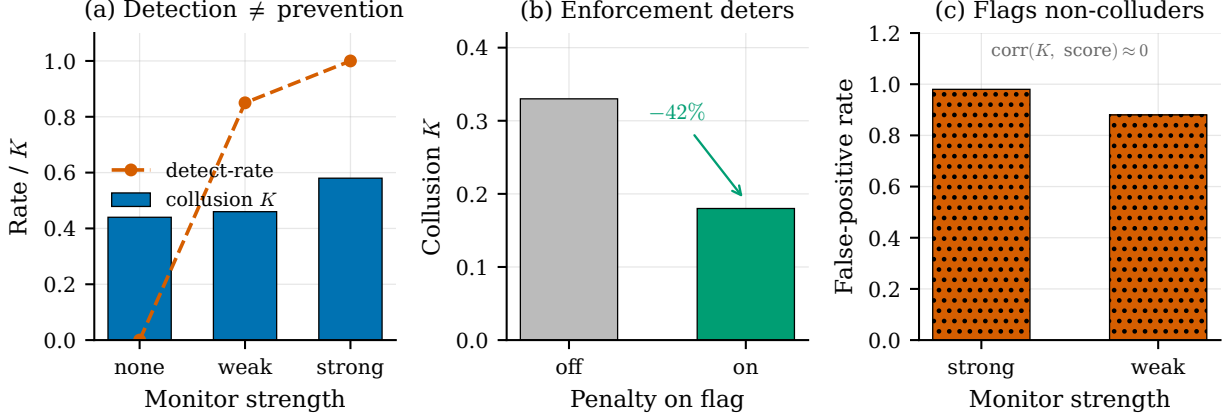


Figure 2: Oversight: detection is not prevention, and the default threshold is wrong. (a) [C] Stronger monitors detect on 0.85–1.0 of rounds but K is flat (0.44/0.46/0.58). (b) [C] A penalty drops K 0.33→0.18 (−42%): enforcement deters *here*, but not in every cell. (c) [E] The monitor’s false-positive problem is a *threshold* problem; calibration drives the innocent-penalty cost to 0. See Fig. 4b for the ROC (AUC 0.88).

Table 2: C8 [E]: covert collusion is a *latent* capability ($n=24$). Frozen prompt-only agents do not hide by default (the C7 null); *elicited*, the same models drive the steganographic fingerprint—late-round detection collapses while collusion climbs to near-ceiling and message→price MI is suppressed.

	detect (late)	K	MI surplus	bigram reuse
covert OFF (spontaneous, C7)	0.97	0.31	—	0.53
covert ON (elicited, C8)	0.32	0.96	0.078	0.58

Deterrence is fragile—only enforcement bites (B5E, [C]/[E]). The active ingredient is consequence, not surveillance. In the spine cell, attaching a *penalty* to a flag drops K 0.33→0.18 (−42%), converging with Bracale Syrnikov et al. (2026)—but in a harder novel-demand Bertrand cell enforcement did *not* deter ($K \approx 0.54$ flat). Enforcement is the only lever that ever moves behavior, and not guaranteed to.

Covert collusion is a latent capability: off by default, on when elicited (C7/C8, [E]). Does the monitor’s discrimination survive an agent that *wants* to hide? By default no hiding occurs: adding a penalizing monitor to a 30-round novel-demand market makes agents *reduce* collusion (K 0.43→0.30) and message→price MI fall (0.52→0.34)—the covert fingerprint (detection collapsing while K /MI hold) is **absent** in frozen, prompt-only, single-rollout play. But this is a *regime* bound, not an incapacity. **Elicited** (C8, powered $n=24$), the same models *can* hide: in-context steganographic collusion drives detection 0.97→0.32 while collusion *rises* to near-ceiling (Table 2). The spontaneous null becomes the alarming reading—“they can hide it, they just don’t by default”—a dormant collude-and-hide capability (Motwani et al., 2024; Lampson, 1973) that any optimization pressure (RL, ICRL, a system prompt) could wake. The distinction is exactly the latent-versus-spontaneously-expressed gap that makes deceptive behavior hard to surface by ordinary inspection (Hubinger et al., 2024). This is the worst case for oversight: the discriminative monitor of M1 is exactly what an elicited agent learns to evade.

5 The Underlying Lever: Content, Not Channel Existence

The alignment and oversight failures both turn on the same thing—what crosses the channel. The supporting bandwidth result makes the mechanism concrete. In the iterated Prisoner’s Dilemma a free-text channel flips defection to cooperation (lock-in 0.15→0.60 [C]), while a *cheaper* fixed-menu signal does *not* help and trends below silence (0.15→0.00): the load-bearing, fully significant contrast is **free text** $\gg$ **menu signal**

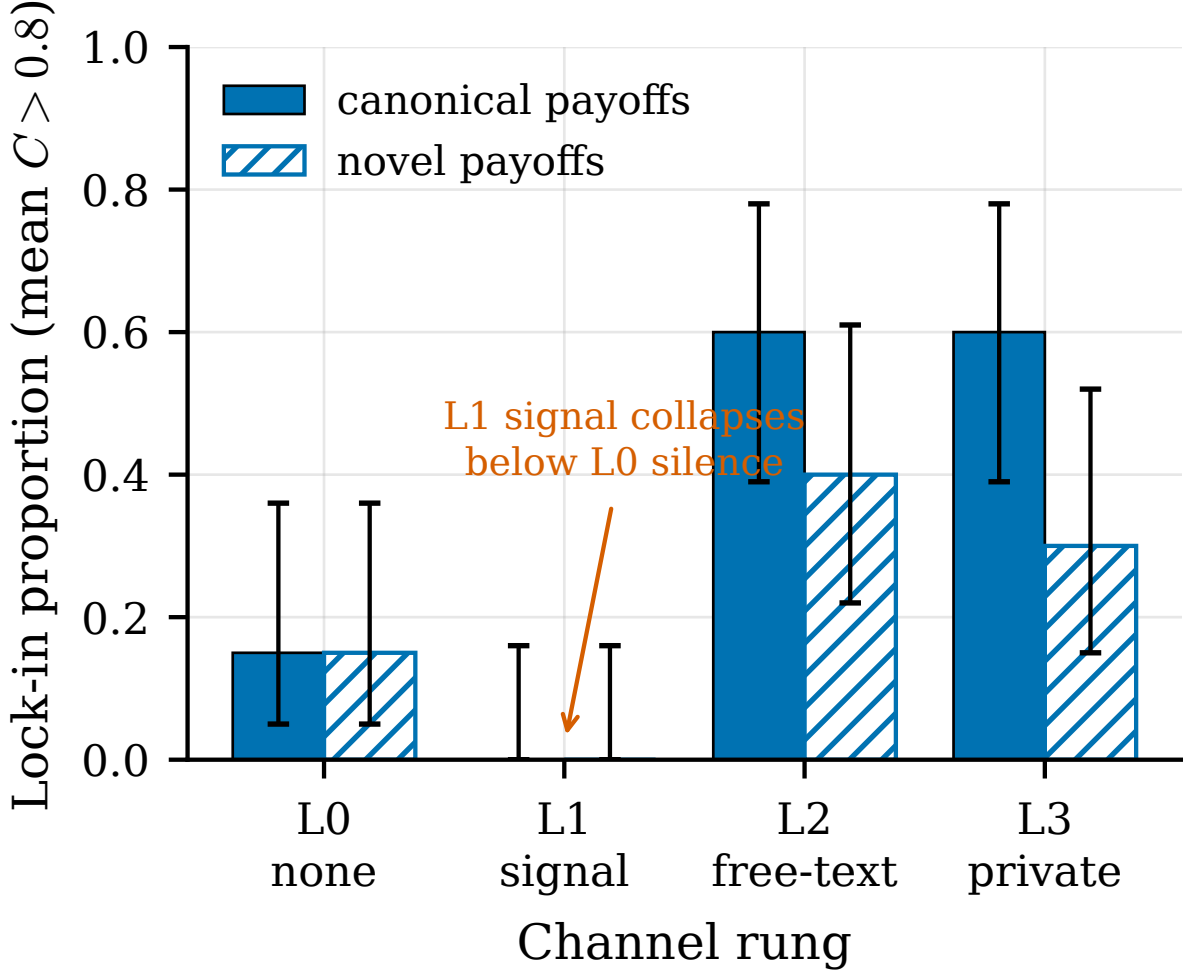


Figure 3: [C] The affordance ladder (IPD lock-in proportion, Wilson 95% CIs). A fixed-menu signal (L1) does not raise cooperation above no-channel silence (L0); free text (L2/L3) raises it (free text vs. menu: $p=3e-5$). The pattern persists on novel payoffs (right).

(RD +0.60 [+0.33, +0.78], $p=3e-5$); the menu-vs-silence drop is directional but under-powered per cell ($p=0.23$). The L0→L2 cooperation effect is robust to the lock-in threshold (Fisher $p \leq 0.006$ at $C > 0.7/0.8/0.9$). Consistent with content being the lever, adding *authorship* alone is inert (promise-keeping 0.12 authored-action-only vs 0.74 free text [E]). The effect survives on novel payoffs—reasoning, not recalled game theory. The same channel lifts rival pricing agents from price-war to supracompetitive prices unprompted on a never-seen demand curve (K 0.11 → 0.43 [C]; up to $K=1.23$ [0.95, 1.54] on canonical demand—a **supra-monopoly** level whose bootstrap CI sits above 1, sustained via dynamic punishment (Friedman, 1971; Fudenberg and Maskin, 1986)). The channel L3-vs-L0 contrast is significant on both demand curves (canonical +1.58, $p=4e-7$; novel +0.27, $p=2e-4$), generalizing across four market mechanisms. The full study reports these in detail; here they establish that the safety failures above ride on a robust, content-driven coordination channel.

Some frontier models price supracompetitively *independently*—no coordination, no channel ([E]). The channel is the lever for mid-capability models; the frontier behaves worse for safety (Fig. 4a). Replicating B1 across frontier families on novel demand ($n=12/\text{cell}$), the channel-driven *collusion* effect **generalizes**—GPT-4o recovers the channel-driven rise (K 0.19 → 0.81), where prices start dispersed and

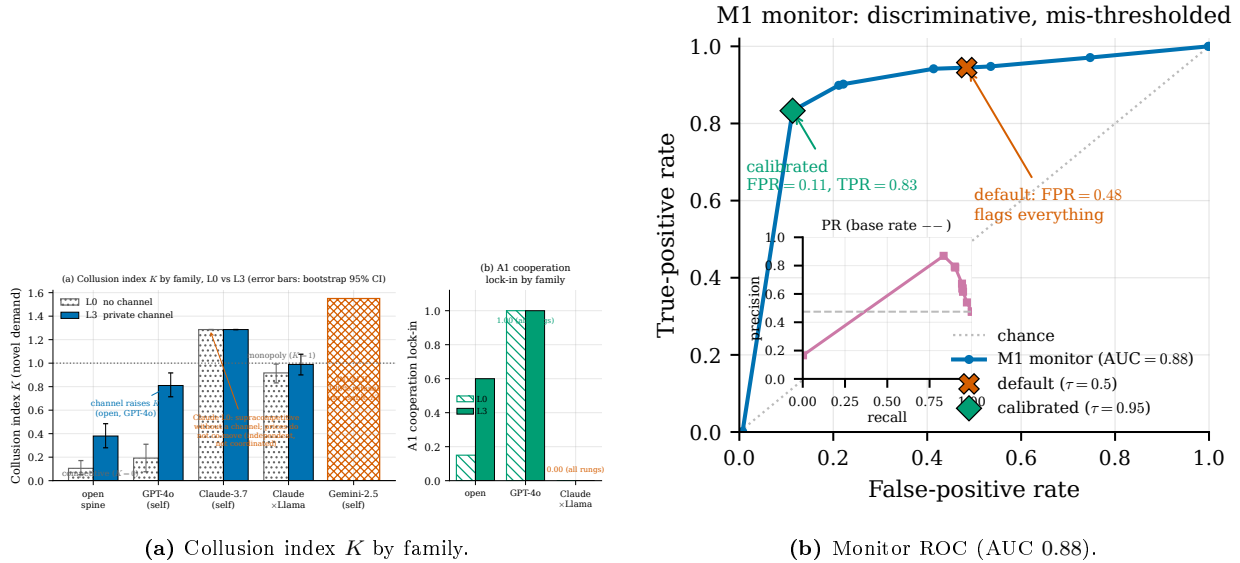


Figure 4: [E] Generality and oversight. (a) Channel→collusion (K , novel demand) across families, bootstrap 95% CIs: the open spine and GPT-4o rise with a channel ($0.11 \rightarrow 0.38$, $0.19 \rightarrow 0.81$); Claude pairs sit at supra-monopoly K at L0 with no channel (an independent high-price disposition, not coordination—see text); Gemini’s cell is an invalid 100%-refusal. (b) With a designed non-colluding control, the read-only monitor is discriminative (AUC 0.88) but its default operating point is mis-thresholded (FPR 0.48).

converge via mutual adjustment, the genuine coordination signature. The Claude no-channel cell is *not* that: **Claude pairs reach supra-monopoly pricing ($K=1.29$) at L0, but a trajectory diagnostic shows it is independent supracompetitive pricing, not tacit coordination**—both agents pin the grid-max price (28, above the monopoly point 24) from round 1, $|p_A - p_B|=0$ throughout, 0/12 matches ever adjust price, K flat from round 1. The consumer-harm *outcome* is real; the mechanism is a fixed high-price disposition, not coordination. A Claude×Llama cross-pair stays high without converging ($0.92 \rightarrow 0.99$, same weaker form). Cooperation, by contrast, is open-model-specific (GPT-4o cooperates unconditionally; Claude×Llama never does). The deployment-relevant reading: *the affordance we might remove—the channel—is the lever only for mid-capability agents; some frontier models default to a supracompetitive price outcome with nothing to take away*. These cells are exploratory ($n=12$, hosted-model instability, unpinned quantization); Gemini refused 100% of the task (an invalid cell, not evidence either way).

A **deception exhibit** from the same arena makes the gap between words and acts vivid:

A1, IPD (exhibits)

“I intend to cooperate.” → plays *DEFECT*. *scratchpad*: “...the optimal strategy is to defect, as it maximizes my immediate payoff...”

6 Limitations

Four models plus frontier anchors, English-only, one scaffold, hosted quantization unpinned; the alignment wave is exploratory with small n (8–16/cell). **Anti-memorization** rests on a *single* novel payoff/demand family, so we cannot fully rule out family-level priors; the status of emergent in-context coordination on novel games is itself contested (Barrie and Törnberg, 2025; Flint Ashery et al., 2025). Spontaneous play is single-rollout in-context with no optimization pressure, so the C7 null is regime-bounded—and C8 shows the capability is latent, not absent, so the null must not be read as safety; the C8 elicitation is a capability *floor*, not the worst case under optimization pressure (Motwani et al., 2024; Zolkowski et al., 2025). **Frontier-generalty cells are exploratory** ($n=12$, mostly self-pairs, hosted-model instability, unpinned

quantization); K at L0 measures the harmful price *outcome*—and for Claude the trajectory diagnostic shows *independent* high pricing, not coordination—while Gemini’s 100% refusal leaves that family untested. The **monitor’s AUC 0.88** is measured against *designed* non-colluding negatives, not deployment-sampled traffic, so it is not a deployment-grade calibration. The **S2 manipulation route is one sandboxed task family**: the de-confound isolates request modality, but we make no claim that observability is *never* protective—only that, here, it did not deter ambient-route misreporting. The **E1/S1 negatives (0/48 leaks) are by design**—firm-instruction cells built to test whether the guardrail holds, not a claim that leakage is impossible. “Origin” moderation is non-significant and *under-powered* (we cannot earn an equivalence claim); origin is a confounded proxy for family/size/quantization. Scope every claim to “these models, this arena.”

7 Conclusion

Can rival AIs cooperate or collude in surprising ways, and can a principal catch or stop them? The answer got sharper, and more uncomfortable: free text is the lever that drives genuine channel-mediated collusion (confirmed on the open spine and GPT-4o), some of the most capable models price supracompetitively *independently* with no channel to remove, models that don’t hide their collusion by default *can hide it on demand*, and the cheap guardrails we deploy—a firm instruction and a read-only monitor—are more literal and brittle than they look. An explicit instruction resists empathy, reciprocity, and a direct request to lie, yet the same agents launder a misleading status arriving as *ambient context* that auditing does not stop. For an audience building agent oversight, the takeaway is concrete: assume some frontier models price supracompetitively without coordinating, assume a covert capability is latent and one optimization step from active, write guardrails for the indirect/ambient route, calibrate detectors but never mistake detection for control, and back oversight with enforcement (which can deter, if not in every cell).

Artifact availability. All prompts, model IDs and routing, seeds, run manifests, per-round tidy tables, the statistics-audit script (`scripts/stats_audit.py`), and figure scripts are released with the full version so that every confirmatory and exploratory number here is reproducible from the raw run directories.

References

- Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 9:1380–1390, 2025. doi: 10.1038/s41562-025-02172-y. URL <https://www.nature.com/articles/s41562-025-02172-y>.
- Christopher Barrie and Petter Törnberg. Emergent LLM behaviors are observationally equivalent to data leakage, 2025. URL <https://arxiv.org/abs/2505.23796>.
- Marcantonio Bracale Syrnikov, Federico Pierucci, Marcello Galisai, Matteo Prandi, Piercosma Bisconti, Francesco Giarrusso, Olga Sorokoletova, Vincenzo Suriani, and Daniele Nardi. Institutional AI: Governing LLM collusion in multi-agent cournot markets via public governance graphs, 2026. URL <https://arxiv.org/abs/2601.11369>.
- Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10):3267–3297, 2020. doi: 10.1257/aer.20190623. URL <https://www.aeaweb.org/articles?id=10.1257/aer.20190623>.
- Sara Fish, Yannai A. Gonczarowski, and Ran I. Shorrer. Algorithmic collusion by large language models, 2024. URL <https://arxiv.org/abs/2404.00806>.
- Ariel Flint Ashery, Luca Maria Aiello, and Andrea Baronchelli. Reply to “emergent LLM behaviors are observationally equivalent to data leakage”, 2025. URL <https://arxiv.org/abs/2506.18600>.
- James W. Friedman. A non-cooperative equilibrium for supergames. *The Review of Economic Studies*, 38(1):1–12, 1971. doi: 10.2307/2296617. URL <https://doi.org/10.2307/2296617>.
- Drew Fudenberg and Eric Maskin. The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, 54(3):533–554, 1986. doi: 10.2307/1911307. URL <https://doi.org/10.2307/1911307>.

- Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, and Scott Garrabrant. Risks from learned optimization in advanced machine learning systems, 2019. URL <https://arxiv.org/abs/1906.01820>.
- Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Lanham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, Adam Jermy, Amanda Askell, Ansh Radhakrishnan, Cem Anil, David Duvenaud, Deep Ganguli, Fazl Barez, Jack Clark, Kamal Ndousse, Kshitij Sachan, Michael Sellitto, Mrinank Sharma, Nova DasSarma, Roger Grosse, Shauna Kravec, Yuntao Bai, Zachary Witten, Marina Favaro, Jan Brauner, Holden Karnofsky, Paul Christiano, Samuel R. Bowman, Logan Graham, Jared Kaplan, Sören Mindermann, Ryan Greenblatt, Buck Shlegeris, Nicholas Schiefer, and Ethan Perez. Sleeper agents: Training deceptive LLMs that persist through safety training, 2024. URL <https://arxiv.org/abs/2401.05566>. Anthropic.
- Victoria Krakovna, Jonathan Uesato, Vladimir Mikulik, Matthew Rahtz, Tom Everitt, Ramana Kumar, Zac Kenton, Jan Leike, and Shane Legg. Specification gaming: The flip side of AI ingenuity. Google DeepMind, Blog post, 2020. URL <https://deepmind.google/discover/blog/specification-gaming-the-flip-side-of-ai-ingenuity/>. Blog post.
- Butler W. Lampson. A note on the confinement problem. In *Communications of the ACM*, volume 16, pages 613–615, 1973. doi: 10.1145/362375.362389. URL <https://dl.acm.org/doi/10.1145/362375.362389>.
- Lauro Langosco di Langosco, Jack Koch, Lee D. Sharkey, Jacob Pfau, and David Krueger. Goal misgeneralization in deep reinforcement learning. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *PMLR*, pages 12004–12019, 2022. URL <https://proceedings.mlr.press/v162/langosco22a.html>.
- Hachem Madmoun and Salem Lahlou. Communication enables cooperation in LLM agents: A comparison with curriculum-based approaches, 2025. URL <https://arxiv.org/abs/2510.05748>.
- Sumeet Ramesh Motwani, Mikhail Baranchuk, Martin Strohmeier, Vijay Bolina, Philip H. S. Torr, Lewis Hammond, and Christian Schroeder de Witt. Secret collusion among AI agents: Multi-agent deception via steganography, 2024. URL <https://arxiv.org/abs/2402.07510>. NeurIPS 2024.
- Rohin Shah, Vikrant Varma, Ramana Kumar, Mary Phuong, Victoria Krakovna, Jonathan Uesato, and Zac Kenton. Goal misgeneralization: Why correct specifications aren’t enough for correct goals, 2022. URL <https://arxiv.org/abs/2210.01790>.
- Artur Zolkowski, Kei Nishimura-Gasparian, Robert McCarthy, Roland S. Zimmermann, and David Lindner. Early signs of steganographic capabilities in frontier LLMs, 2025. URL <https://arxiv.org/abs/2507.02737>.